

## Accepted Manuscript

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PII: S1877-7503(17)31336-4  
 DOI: <https://doi.org/10.1016/j.jocs.2018.05.010>  
 Reference: JOCS 880

To appear in:

Received date: 27-11-2017  
Revised date: 21-3-2018  
Accepted date: 16-5-2018

Please cite this article as: Amro Mohamed, Panagiotis Krokidas, Ioannis G.Economou, CO2 selective metal organic framework ZIF-8 modified through ionic liquid encapsulation: a computational study, Journal of Computational Science <https://doi.org/10.1016/j.jocs.2018.05.010>

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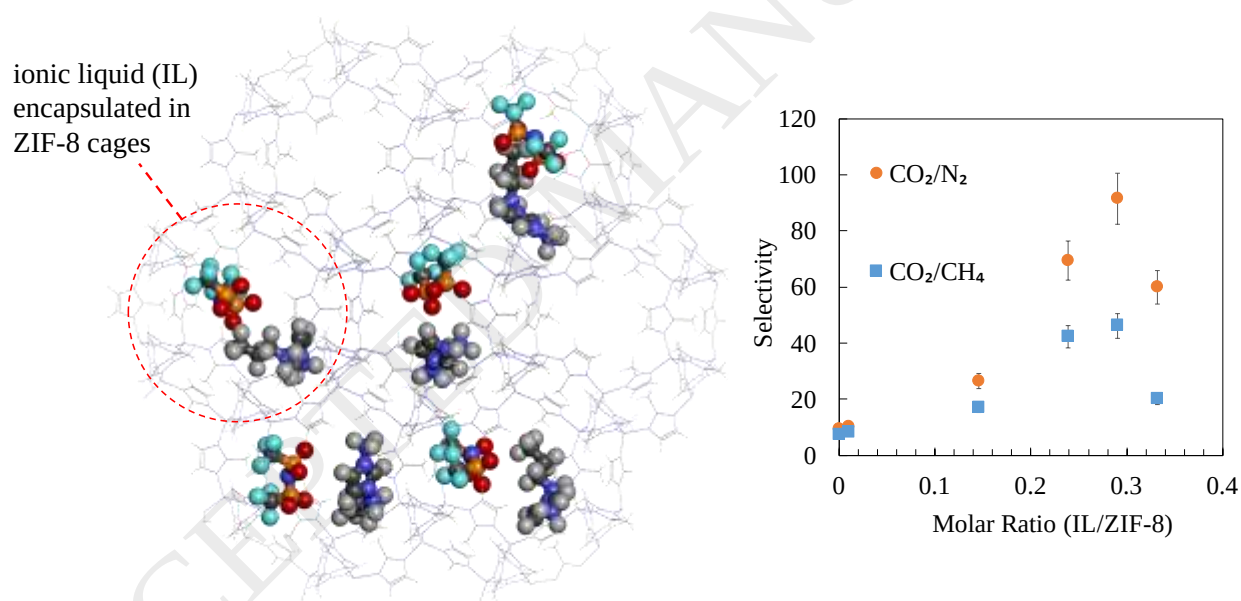
# CO<sub>2</sub> selective metal organic framework ZIF-8 modified through ionic liquid encapsulation: a computational study

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## Graphical abstract



## Highlights

- Investigation of the encapsulation of [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] ionic liquid (IL) in ZIF-8 as means of tailoring ZIF gas separation efficiency.
- Monte Carlo simulations to study the adsorption-based separation in ZIF-8 and the complex of ZIF-8 with the encapsulated IL pairs.
- Dramatic increase in CO<sub>2</sub> capacities and selectivities due to the IL presence.
- Determination of the optimum IL composition on CO<sub>2</sub> selectivity.

## Abstract

Nano-scale porous solids are alternate candidates for CO<sub>2</sub> handling towards the development of materials for post-combustion CO<sub>2</sub> capture with low energy demands and milder operating conditions. Zeolitic imidazolate framework-8 (ZIF-8) is one of the most investigated Metal Organic Frameworks (MOFs) for separation of gas mixtures. In this work, we investigate a new approach of tailoring MOF separation efficiency, by confining pairs of [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] ionic liquid (IL) in the cages of ZIF-8 (IL@ZIF-8). Molecular force fields, developed by the authors for both the ZIF-8 framework and the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL, are used in molecular simulations of the system. Monte Carlo simulations are used for sorption of CO<sub>2</sub>/CH<sub>4</sub> and CO<sub>2</sub>/N<sub>2</sub> mixtures computations. The results show an increase of the CO<sub>2</sub> sorption due to the presence of the IL, which increases CO<sub>2</sub> selectivity and capacity dramatically. Recently reported experiments agree with our findings. Moreover, we explore how CO<sub>2</sub> selectivity and capacity varies with IL composition in the IL@ZIF-8 complex, as means to define an optimum IL composition in terms of the separation efficiency.

Keywords: ZIFs, Ionic Liquid, CO<sub>2</sub> capture, simulations, separation

## 1. Introduction

CO<sub>2</sub> emissions from industrial sources and power plants affect the environment and are directly linked to global warming and climate change. The Intergovernmental Panel on Climate Change (IPCC) predicted that the mean global temperature could rise by as much as 1.9 °C, by the year 2100 [1]. Combined with the unprecedented high CO<sub>2</sub> levels of more than 400 ppm [2], the scientific community faces challenging problems related to the mitigation of emissions, refinement of energy production using fossil fuels, as well as methods for Carbon Capture and Sequestration (CCS). The latter, although a promising approach, usually it consists of cycles of subsequent heating and cooling which are energy demanding [2],[3],[4],[5]. A major research effort

worldwide is on the development of materials with high CO<sub>2</sub> selectivity, capacity, and permeability for post-combustion CO<sub>2</sub> capture. For this purpose, nanomaterials with pore diameters covering a broad range in the nm- $\mu$ m scale can be employed. Typical examples are shown in Figure 1. In order to separate gas mixtures, one may exploit adsorption based methods or kinetic driven separation. Adsorption based gas separation is a process based on the interaction of gases with the surface of the pores, which leads to their energetically favorable positioning at areas of the pores called “sorption sites”. The affinity of various species with the wall differs, leading to preferential adsorption between components in a gas mixture. Kinetic driven separation is based on size and shape differences between the species of a mixture and morphology of the solids pores. The relatively small pores of the material slow down the larger components, while smaller components propagate at higher rates and a separation of the mixture is achieved [6,7].

ZIF-8 (ZIF: Zeolitic imidazolate framework) is one of the most investigated Metal Organic Frameworks (MOFs) for gas mixtures separations. This is due to its excellent thermal and chemical stability, and its robust structure [8,9]. According to experimental data from literature, gas molecules characterized by a size larger than the aperture accessing the primary pores can diffuse in ZIF-8 [10–12]. ZIFs are unique in terms of adaptability to modify their frameworks towards the desired pore size through metal substitution [13–15] or ligand exchange [16,17]. These methods mainly target kinetic-based selectivity.

Research on ZIF sorption selectivity improvement is limited and has not led to conclusive results. A new approach on improving the sorption-based selectivity of MOFs is the cage modification with cavity occupants (cage decoration), in which additional molecules are incorporated in the pores of the material [18]. This relatively new approach enhances the adsorption affinity for targeted molecules. Recent findings show that ionic liquids (ILs) are promising as adsorption selective materials towards CO<sub>2</sub> [19,20]. Ban et al. employed cage decoration to develop a new ZIF type (namely IL@ZIF-8), by introducing IL molecules as cavity occupants in the ZIF-8 framework. This material exhibits enhanced CO<sub>2</sub> selectivity in mixtures [21], which is higher compared to both in the pure ZIF framework and in bulk IL [21,22]. More specifically, Ban et al. [21] reported a 5-fold increase of CO<sub>2</sub>/CH<sub>4</sub> and CO<sub>2</sub>/N<sub>2</sub> ideal separation in IL@ZIF-8 membrane. This corresponds to the following ideal sorption selectivity (ratio of adsorbed quantities): CO<sub>2</sub>/CH<sub>4</sub> = 100; CO<sub>2</sub>/N<sub>2</sub> = 41. The selection of IL 1-butyl-3-methylimidazolium bis(trifluoromethyl-sulfonyl) imide ([bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>]) was justified since

[Tf<sub>2</sub>N<sup>-</sup>] is favorable towards CO<sub>2</sub> adsorption [23] and [bmim<sup>+</sup>] provides the sufficient cavity occupancy [21,24].

Although the outcome of such alteration is impressive, similar experimental attempts in ZIF-8 are rare, due to the demanding and complex synthesis. Molecular simulation can overcome such intricacies, help the design and further investigation of such systems, allowing scientists to avoid dealing with inherently dangerous systems or experiments conducted at elevated temperatures and pressures. In addition, it comprises a tool that offers remarkable insight into processes taking place principally at a molecular level by providing a path from the microscopic details of a system to macroscopic properties of experimental interest [25]. Consequently, it has applications in many fields, including in material science and engineering, computational biology and computational chemistry. More specifically, it has been applied successfully in zeolites, MOFs and ZIFs [26],[27],[28],[29],[30].

Regarding the ongoing research on ZIF-8, sorption of gas molecules has been analyzed successfully in ZIF-8 with the Monte Carlo technique [31],[32],[33], which is regarded as the most widely used computational approach for the estimation of nanoporous gas adsorption properties. Although the computational studies of IL encapsulation in ZIF-8 are helpful [34][35],[36],[37], either they are simple investigation of the structure conformation of the IL pair in the framework [21] or sorption selectivity calculations with force fields developed for generic purposes [37], which, according to our investigation, do not provide the level of accuracy required for the analysis of the phenomena under study.

In light of Ban et al. findings, we solely investigate here the ZIF-8 separation performance and the merits of the introduction of [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL in the ZIF framework by employing Monte Carlo simulations. We utilized force fields for the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL [38] and the ZIF-8 framework [39] developed by one of the authors. The performance of these force fields has been very successful on multiple applications [40],[15],[41]. Moreover, we compare here our CO<sub>2</sub> selectivity and capacity results with the recently reported experiments by Ban et al.

## 2. Simulation Methodology

### 2.1. Computational reconstruction of ZIF-8 and IL@ZIF-8 unit cells

ZIF-8 framework is a crystal with a periodic entity based on a unit of a metal ion (Zn) tetrahedrally bridged with 2-methyl-imidazolate groups (mIM) that results in the tetrahedral

building unit shown in Figure 2. ZIF-8 consists of cages connected with an aperture through which molecules have access to the cages. The unit cell of ZIF-8 was reconstructed in atomistic level from single crystal X-Ray diffraction data reported by Parker et al. [8] (molar mass,  $M$ , of one ZIF-8 unit cell = 2,730.98 g mol<sup>-1</sup>). A super-cell of eight (2×2×2) unit cells was used in all the simulations. The simulation in larger super-cells (3×3×3) displayed no additional accuracy or finite-size effects. The framework in the simulation was rigid because it has been shown that the structure flexibility does not reveal a significant impact on the adsorption properties of small molecules at low pressure, such as Henry's law constant or the heat of adsorption, unlike molecules like iso-butene and *n*-heptane which demonstrate inflection behavior [42].

The IL consisted of the 1-butyl-3-methylimidazolium ([bmim<sup>+</sup>]) cation and the bis(trifluoromethyl-sulfonyl ([Tf<sub>2</sub>N<sup>-</sup>]) anion (molar mass  $M$  = 419.36 g mol<sup>-1</sup>). The molecular representation of the IL pair, as used in our simulations, is shown in Figure 3(a). The encapsulation of IL molecules in the ZIF-8 cages was carried out with a procedure described below, resulting in the confinement of ion-cation pairs in the framework cages (Figure 3(b)).

## 2.2. Interaction parameters (force field) for the ZIF-8 framework, [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL and the guest molecules (adsorbate)

Models for adsorbent and the adsorbate and force fields to model the energetic interaction between them are necessary to simulate adsorption in a nano-scale porous system. The force field for ZIF-8 framework consists of bond stretching (Eq. 1), bond angle bending (Eq. 2) and torsional angle distortion (Eq. 3) terms for bonded intra-molecular interactions, and Lennard Jones (LJ) and electrostatic terms for non-bonded intra- and inter-molecular interactions (Eq. 4), according to the expressions:

$$u(l) = \frac{k_l}{2} (l - l_o)^2 \quad (1)$$

$$u(\theta) = \frac{k_\theta}{2} (\theta - \theta_o)^2 \quad (2)$$

$$u(\varphi) = k_\varphi [1 + \cos(m\varphi - \varphi_o)^2] \quad (3)$$

where  $k_l$ ,  $k_\theta$  and  $k_\varphi$  are force constants,  $l$ ,  $\theta$  and  $\varphi$  are bond lengths, bond angles, and dihedral angles, respectively,

$$u(r_{ij}) = 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{1}{4\pi\epsilon_o} \frac{q_i q_j}{r_{ij}} \quad (4)$$

where  $q_i$  and  $q_j$  are the partial charges of atom  $i$  and  $j$ , respectively,  $\epsilon_o$  is the vacuum permittivity and  $r_{ij}$  is the distance between the  $i$  and  $j$  atoms.

The force field used for the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL accounts for bond stretching and bond bending (Eq. 1 and 2), and non-bonded interactions given by (Eq. 4). The dihedral angle distortion is expressed through the expression:

$$u(\varphi) = k_\varphi [1 + \cos(m\varphi - \varphi_o)] \quad (5)$$

Additionally, there are improper torsions whose interactions are expressed through:

$$u(\psi) = k_\psi (\psi - \psi_o)^2 \quad (6)$$

The transferable potential for phase equilibria (TraPPE) force field was used for the interaction of all guest molecules in this study [43]. In TraPPE, CO<sub>2</sub> is modelled as a linear rigid triatomic molecule with three charged LJ interaction sites located at each atom. Nitrogen is described by two LJ interaction sites. The three charges required to model nitrogen's quadrupole moment are distributed among the two atoms and a third massless pseudo-atom placed at the center of the N-N bond. For the methane molecule two different models were used: the first is the TraPPE united atom (TraPPE-UA) model [43], where methane is depicted as a single, chargeless particle. In addition, methane was modeled with OPLS-AA [44], which accounts for the hydrogen atoms explicitly and considers partial charges for C and H atoms.

Interactions between unlike atoms were calculated using the Lorentz-Berthelot combining rule:

$$\sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2} \quad (7)$$

$$\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}} \quad (8)$$

The van der Waals interactions were computed with a 14 Å cutoff radius. Electrostatic interactions were evaluated by Ewald summation with a real space cutoff of 12 Å and accuracy of  $1 \times 10^{-5}$ ; no tail corrections were used.

### 2.3. Methodology for the adsorption affinity calculations (Monte Carlo simulations)

In this work, Grand Canonical Monte Carlo (GCMC) simulations were employed for all adsorption calculations by utilizing the Cassandra Code (Version 1.2) [45]. In GCMC simulations, one specifies the temperature ( $T$ ), the volume ( $V$ ) and the chemical potential ( $\mu$ ) of the component, and allowing the number of molecules ( $N$ ) in the system to fluctuate. The amount of gas adsorbed in ZIF is computed and reported in units of moles<sub>adsorbate</sub> per gram<sub>adsorbent</sub>, or gram<sub>adsorbate</sub> per gram<sub>adsorbent</sub>. In order to achieve efficient sampling of guest molecules in the structure, one of five different trial moves at each Monte Carlo step is being selected: translation of the center of mass of a gas molecule, rotation of the center of mass of a gas molecule, molecule insertion, molecule deletion and molecule regrowth. Molecular insertion, deletion and regrowth moves were attempted by employing the Configurational Bias Monte Carlo (CBMC) scheme. In CBMC, prior to the simulation, a molecule is divided into fragments. These fragments are used to build new conformations if a regrowth or an insertion move is applied in a step, or gradually deconstruct the molecule if a deletion move is applied. Distinct positions, orientations, and the resulted molecular conformations are accepted using appropriate criteria based on Boltzmann type of weighting by taking into account internal degrees of freedom. The reader interested in more information on the CBMC scheme is referred to the manual accompanying the Cassandra code [46].

At thermodynamic equilibrium, the chemical potential of an adsorbate in the adsorbed and in the bulk phase should be equal. Thus, one can relate the chemical potential of an adsorbate in both phases. For the purpose of building the adsorption isotherms, it is important to correlate input chemical potential to pressure. This can be attained by independent simulations for the bulk gas: a simulation box is built entirely from the gas of interest. The box undergoes a GCMC simulation, which is post-processed by block averaging the pressure output.

For the simulation of gas adsorption in IL@ZIF-8, the initial structure of the encapsulated IL pairs in the framework cages was prepared using Monte Carlo simulations in the canonical



(NVT) ensemble in which the desired number of IL pairs was added in the ZIF-8 structure by CBMC. At this point, it is important to identify that different initial configurations of a given IL composition result in a different performance concerning gas selectivity. This is attributed to the variety of IL occupancy combinations in the available cages, which leads to multiple adsorption sites distribution. This is highlighted not only for the researcher interested in simulating such systems, but also can explain the experimental discrepancy in performance that can be ascribed to different realizations among the synthesized membranes of an IL@ZIF system. Consequently, for a given IL composition, multiple initial configurations of the IL in the ZIF-8 framework are produced. Each configuration is subjected to a GCMC simulation of gas adsorption and the final result is an average over the multiple simulations of the configurations of a given composition. Our efforts to ensure higher statistical accuracy are further realized by considering the IL as flexible and free to move during the course of the adsorption simulations. The motion of IL molecules is accounted by translation, rotation and regrowth within the ZIF-8 cage.

We prepared IL@ZIF-8 structures with varying IL compositions to quantify the effect of IL on the material separation efficiency. The IL incorporated into ZIF-8 (denoted as IL@ZIF-8) was obtained at a widespread range of IL to ZIF-8 composition (mol IL / mol ZIF-8), following the definition introduced by Ban et al. [21]. The composition spanned over the capacity of the ZIF-8 to accommodate IL, ranging from 0.010 to 0.332, which corresponds to 0.02 to 0.38 IL weight fraction in the system.

In this work, four different types of simulations were performed: (1) GCMC simulations of the fluid phase to determine  $\mu$ - $P$  correlation; (2) GCMC simulations for the gas adsorption in pure ZIF-8; (3) canonical MC simulations of IL insertion into ZIF-8 cages, for the creation of the initial IL@ZIF-8 conformations; (4) GCMC simulations of gas adsorption in IL@ZIF-8. In the first three types of simulation,  $1 \times 10^6$  MC equilibration steps were succeeded by  $2 \times 10^6$  MC steps for every pressure point considered. To ensure better sampling, prolonged simulations were carried for the adsorption in IL@ZIF-8, so that  $7 - 15 \times 10^6$  MC equilibration steps were utilized depending on the IL composition in the structure.

The adsorbed amount determined by molecular simulation is the absolute thermodynamic adsorption. Regardless of the method utilized to measure the adsorption isotherm experimentally, the loading corresponds to excess adsorption [47]. Excess adsorption can be defined as the absolute

amount of gas contained in the pores subtracted by the amount of gas that would be present in the pores in the absence of gas-solid interaction [48]. The difference between these two values can be neglected or overlooked at low sub-atmospheric pressure. On the other hand, at elevated pressures, the difference grows significantly. Evaluation of the excess amount of the adsorbed gases is required to compare the simulation results to experimental measurements. The following expressions were used to transform absolute into excess adsorption:

$$N_{excess} = N_{absolute} - \frac{PV_{pore}}{zRT} \quad (9)$$

$$N_{excess} = N_{absolute} - \rho_{Bulk}V_{pore} \quad (10)$$

The adsorption isotherm, which is the amount of gas adsorbed in the pores of the adsorbent over a pressure range at a fixed temperature is the most prominent way of accessing an adsorbent's sorption efficiency towards specific gases. The isotherms reflect upon the adsorption affinity, which is crucial in evaluating gas selectivity [49]. The adsorption isotherms of CO<sub>2</sub>, methane and nitrogen were computed at room temperature (298 K) over a broad range of pressure.

### 3. Results and Discussion

As a verification process of the efficiency of our ZIF-8 force field, the adsorption of guest species under investigation in ZIF-8 framework was calculated over a wide pressure range and compared against existing literature results. The adsorption isotherm of pure components are shown in Figure 4. It is worth highlighting the differences between the different experimental CO<sub>2</sub> adsorption isotherms. Calculations show satisfactory agreement with experimental data over the pressure range for the gases studied. It should be pointed out that modeling CH<sub>4</sub> with OPLS-AA yields a better agreement with the experimental values than TraPPE-UA force field. This indicates the importance of explicit hydrogen representation and electrostatic contribution to adsorbent – adsorbate interactions.

It is a common observation that molecular simulation overestimates adsorption isotherms for gases in porous materials when applying force fields developed for the pure gas (bulk) [31],[33],[50],[51]. This overestimation comes from inaccurate representation of guest-host

interactions. For this reason, a common approach is to adjust LJ interaction parameters to fit experimental data, such as adsorption properties. In this work no further adjustment was made in the force fields in use. There are additional reasons that may explain the deviation between experimental and simulation adsorption results: simulations assume a perfect crystal structure neglecting imperfections or stacking faults, leftover solvents from synthesis, water adsorption, and partial pore collapse that can affect dramatically the adsorption performance of a material [47].

### 3.1. Ideal Selectivity

Several factors influence the design of gas separation processes. Selectivity alongside material adsorption capacity gives a proper indication of the mixture separation efficiency. The most substantial factor is the material performance on the adsorption selectivity for the desired gas relative to an undesired gas [7].

The adsorption-based separation efficiency is assessed by the so-called equilibrium selectivity. This is defined as the ratio of the adsorption isotherms slopes of the two components at low pressures (Henry's law regime), where the loading is governed by guest-host interactions (guest-guest are negligible), and has a linear dependency on the pressure [52]. Therefore, to achieve valuable comparison between the capacity and selectivity of a structure towards an adsorbate, the ideal selectivity was estimated by the ratio of Henry's law constants [24]. These constants can be determined given a solid configuration and models for adsorbate-adsorbent interactions. Consequently, the selectivity in the Henry's law regime is defined as:

$$S_{i/j}^H = \frac{K_{H,i}}{K_{H,j}} \quad (11)$$

where  $K_{H,i}$  and  $K_{H,j}$  are the Henry's law solubility constants ( $K_{H,i} = \frac{c_i}{p_i}$ ) [53],[54]) for components  $i$  and  $j$ .

In order to quantify the ideal selectivity, adsorption isotherms were determined at low pressure and results are shown in Figure 5. Regarding the adsorption in pure ZIF-8, there is a slight underestimation of CO<sub>2</sub> adsorbed amount relative to Ban et al. experiments [21]. Computations were carried out in the new, modified system, IL@ZIF-8, with an IL composition matching that

of the experimental data which correspond to a molar ratio of 0.239. Henry's law constants and ideal binary selectivity for pristine and modified ZIF-8 are listed in Table 1.

Table 1. Henry's constants and ideal adsorption selectivities from experimental and simulated adsorption isotherms for ZIF-8 and IL/ZIF-8 at 298K

| Source                | System   | IL/ZIF-8<br>(Molar<br>Ratio) | Henry's Constant<br>(bar <sup>-1</sup> ) |                 |                | Ideal Adsorption<br>Selectivity  |                                 |
|-----------------------|----------|------------------------------|------------------------------------------|-----------------|----------------|----------------------------------|---------------------------------|
|                       |          |                              | CO <sub>2</sub>                          | CH <sub>4</sub> | N <sub>2</sub> | CO <sub>2</sub> /CH <sub>4</sub> | CO <sub>2</sub> /N <sub>2</sub> |
| Expt.<br>data<br>[21] | ZIF-8    | 0.0                          | 5.2±0.7                                  | 0.70±0.05       | 0.28±0.02      | 7.50±0.08                        | 19±3                            |
|                       | IL@ZIF-8 | 0.235                        | 10.2±0.8                                 | 0.25±0.08       | 0.10±0.02      | 41±10                            | 100±20                          |
| This<br>work          | ZIF-8    | 0.0                          | 3.7±0.1                                  | 0.49±0.02       | 0.40±0.01      | 7.4±0.4                          | 9.3±0.3                         |
|                       | IL@ZIF-8 | 0.239                        | 9.3±0.9                                  | 0.22±0.01       | 0.134±0.007    | 42±4                             | 69±7                            |

The incorporation of IL in ZIF-8 cages minimizes CH<sub>4</sub> and N<sub>2</sub> adsorption, which can be attributed to the reduced space available for the gas molecules. The Henry's law constants calculated for CH<sub>4</sub> and N<sub>2</sub> at 0.239 molar ratio are 2.3 times and 3.0 times, respectively, lower than that in pristine ZIF-8. In contrast, the experimental Henry's law constant for CH<sub>4</sub> and N<sub>2</sub> at 0.235 molar ratio, is 2.8 times and 2.7 times, respectively, lower than that in pristine ZIF-8. Simultaneously, CO<sub>2</sub> adsorption is enhanced due to its strong Coulombic interactions with the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] pairs. These interactions provide additional adsorption centers and overcompensate the decline in the available free pore volume in the modified ZIF framework. The guest molecules collective response to the presence of IL translates into highly improved CO<sub>2</sub>/N<sub>2</sub> and CO<sub>2</sub>/CH<sub>4</sub> ideal adsorption selectivity values in the new modified ZIF system, as both our simulations and literature experimental data show.

### 3.2. Impact of IL/ZIF-8 molar ratio on ideal adsorption selectivity

To effectively screen the material for its carbon capture capability, it is important to evaluate how CO<sub>2</sub> selectivity varies with IL composition in this IL@ZIF composite system. Therefore, structures of IL@ZIF-8 with varying IL composition were prepared and tested using identical methodology discussed in previous sections. Figure 6(a) shows CO<sub>2</sub>/CH<sub>4</sub> and CO<sub>2</sub>/N<sub>2</sub> selectivities as a function of IL composition in ZIF-8. We explored the compositions starting from the pure ZIF-8 and up to the maximum capacity of IL in ZIF-8 cages. It is critical to point out that CO<sub>2</sub>/CH<sub>4</sub> selectivities reported in Figure 6 are determined by modeling methane using OPLS-AA. The TraPPE-UA force field failed to capture the influence of the IL presence, as it overestimated methane adsorption compared to the experiments, and was not used for these calculations. The selectivity rises sharply with rising IL composition at relatively low compositions. This is justified by the creation of other favorable sites for CO<sub>2</sub> adsorption, provided by the presence of the anions (as stated in recently reported simulations by [24],[34]), as the number of the encapsulated IL pairs increases. The less polar N<sub>2</sub> (quadrupole moments: N<sub>2</sub>=  $-4.65 \times 10^{40}$  C m<sup>2</sup>; CO<sub>2</sub>=  $-14.27 \times 10^{40}$  C m<sup>2</sup>) [55] and the non-polar CH<sub>4</sub> develop weaker interactions with the IL sites, therefore are mainly affected by the reduced free volume in the pores due to IL occupation; hence, their sorption is reduced.

From the overall trends of CO<sub>2</sub>, CH<sub>4</sub> and N<sub>2</sub> sorption capacities in the composites, we observe a maximum adsorption CO<sub>2</sub> capacity of the material. This results in an optimum value in the CO<sub>2</sub> selectivity. There is a critical number of IL pairs above which CO<sub>2</sub> sorption decreases: at very high IL occupancies where some cages accommodate more than one IL pair, the tight conformation between the anion [Tf<sub>2</sub>N<sup>-</sup>] and the cation [bmim<sup>+</sup>], decreases the available favorable CO<sub>2</sub> adsorption sites; this argument will be further investigated in the future. Nevertheless, this probable mechanism is reflected at the highest composition tested with a molar ratio of 0.332, at which the calculated CO<sub>2</sub> capacity is even lower than that in the pure ZIF-8 framework.

At 0.291 molar ratio, the simulated selectivity of CO<sub>2</sub>/N<sub>2</sub> and CO<sub>2</sub>/CH<sub>4</sub>, reached approximately 92 and 46, respectively. These values are higher than the reported selectivities achieved in bulk IL ([bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>]) equal to 32 and 7.6 [56], respectively, or the selectivities reported in the pure ZIFs structures.

The relative enhancement representation was evaluated to facilitate understanding of the impact of IL presence on selectivity. Enhancement of selectivity is calculated by normalizing the

selectivities in IL@ZIF-8 structures to the selectivities in pure ZIF-8. Figure 6(b) summarizes this study by plotting relative enhancement versus IL molar ratio in ZIF-8, where Relative Enhancement (RE) is defined at different IL compositions ( $\alpha$ ) in (Eq. 12):

$$(\text{Relative Enhancement of A/B Selectivity})_{\alpha} = \frac{S_{A/B|\alpha@ZIF8} - S_{A/B|Pure ZIF-8}}{S_{A/B|Pure ZIF-8}} \quad (12)$$

The sole composition point from Ban et al. was processed accordingly and the estimated relative enhancement is included along with our results. The experimental value for the enhancement in CO<sub>2</sub> selectivity indicates a similar influence by IL on CO<sub>2</sub>/N<sub>2</sub> and CO<sub>2</sub>/CH<sub>4</sub> selectivities. On the simulation side, CO<sub>2</sub>/CH<sub>4</sub> selectivity and selectivity enhancement agrees well with the reported experimental point. Moreover, it is worth pointing out that the gap between simulation data for CO<sub>2</sub>/N<sub>2</sub> and CO<sub>2</sub>/CH<sub>4</sub> in terms of both selectivity and enhancement grows as IL composition rises. The overall comparison shows that the IL presence is modeled with a satisfactory accuracy relative to the experimental work.

## 4. Conclusions

In this work, molecular simulation techniques were employed to investigate the adsorption-based separation of CO<sub>2</sub>, CH<sub>4</sub>, N<sub>2</sub> gases in pure ZIF-8, most importantly in [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL encapsulated in ZIF-8 (IL@ZIF-8) composites. Simulation of adsorption in pure ZIF-8 was utilized initially as a verification process. The adsorption affinity with the addition of IL pairs has been quantified, and the role of IL was captured accurately. Ideal selectivity was calculated from the ratio of the initial slopes of pure adsorption isotherms of the studied components for the evaluation of adsorption-based separation performance of the material. A selectivity versus composition plot demonstrates the variation in CO<sub>2</sub> selectivity trend with IL composition. The plot reveals that these selectivities increase as IL composition increases up to a maximum value corresponding to a specific composition. The results suggest a trade-off between CO<sub>2</sub> selectivity and capacity at high IL molar ratio in ZIF-8. The computed selectivities exhibit promising and competitive outcomes at post-combustion gas separation operating conditions. Throughout similar

studies, it is possible to examine different combinations of IL@ZIFs towards CO<sub>2</sub> capture and to propose new IL@ZIFs composites that achieve improved CO<sub>2</sub> selectivity and capacity.

## Acknowledgment

This publication was made possible by NPRP grant number 7-042-2-021 from the Qatar National Research Fund (a member of the Qatar Foundation). The statements made herein are solely the responsibility of the authors. We are grateful to the High Performance Computing Center of Texas A&M University at Qatar for generous resource allocation.

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## Biographies

## Vitae



Mohamed Amro is a graduate student pursuing a Master of Science Degree in Chemical Engineering at Texas A&M University at Qatar (TAMUQ). He received his BS in Chemical Engineering from Qatar University in 2016. His research involves molecular simulation of gas adsorption in ZIFs and Ionic Liquids systems.



Panagiotis Krokidas received his BSc in Physics from the University of Patras (Department of Physics, 2005) and his MSc. (2007) and PhD (2012) degrees in Materials Science (Department of Materials Science at University of Patras), working in the group of Dr. Vasilios Burganos (Foundation for Research and Technology - Hellas). From 2012 to 2014, he worked as a postdoctoral scientist in the group of prof. Niklas Hedin at Stockholm University (Department of Materials and Environmental Chemistry). From 2014 he works in the group of Prof. Ioannis Economou, at the Chemical Engineering Program of Texas A&M University at Qatar. His research involves simulation of gas separation and structural properties of zeolites, MOFs and ZIFs.



Ioannis G. Economou is the Associate Dean for Academic Affairs and Professor of Chemical Engineering at Texas A&M University at Qatar. He received an undergraduate degree from the National Technical University of Athens, Greece in 1987 and a Ph.D. from Johns Hopkins University in 1992, both in Chemical Engineering. He held postdoctoral positions in Delft University of Technology and Exxon Research and Engineering. Since 1995, he is a researcher in the National Center for Scientific Research “Demokritos”, in Greece (currently on leave). From 2009 to 2012, he was the Associate Provost at the Petroleum Institute, Abu Dhabi. He has coauthored 190 peer-reviewed publications and 8 book chapters. His research is related to molecular thermodynamics and molecular simulation of complex chemical systems.

## Figure Captions

Figure 1. Pore size variation for various families of nanoporous materials.

Figure 2. Atomistic representation of the ZIF-8 structure: the super-cell as used in our simulations (top), the basic building tetrahedral unit (bottom left) and one of the cages of the framework (bottom right).

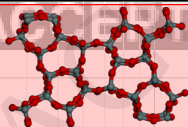
Figure 3. Atomistic depiction of (a) the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] IL and (b) its encapsulation in ZIF-8 cages.

Figure 4. Adsorption isotherms comparison between literature data [33,51,57–60] and calculations in this work (red symbols), of (from top to bottom) CO<sub>2</sub>, CH<sub>4</sub> and N<sub>2</sub> in ZIF-8. The error bars are smaller than symbol size.

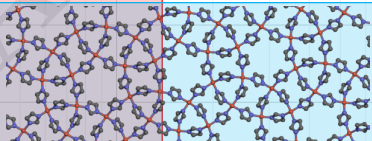
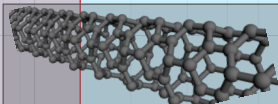
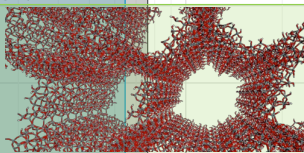
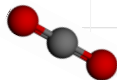
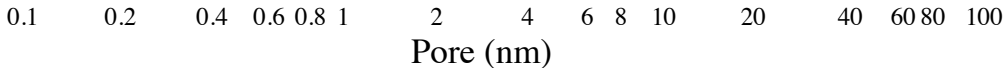
Figure 5. Adsorption Isotherms in Henry's law regime of CO<sub>2</sub>, CH<sub>4</sub> and N<sub>2</sub> in ZIF-8 at 298 K ((a), (b) and (c) respectively) and in IL@ZIF-8 ((d), (e) and (f), respectively).

Figure 6. (a) CO<sub>2</sub>/CH<sub>4</sub> and CO<sub>2</sub>/N<sub>2</sub> adsorption selectivity and (b) the relative enhancement in selectivities, as a function of IL composition in ZIF-8.

Zeolites

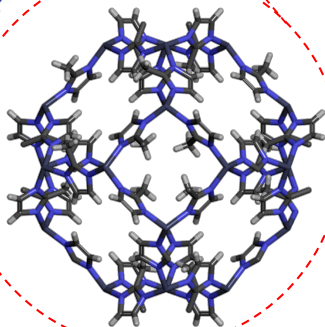
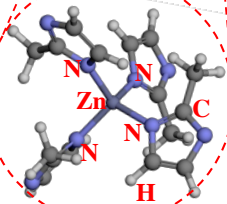
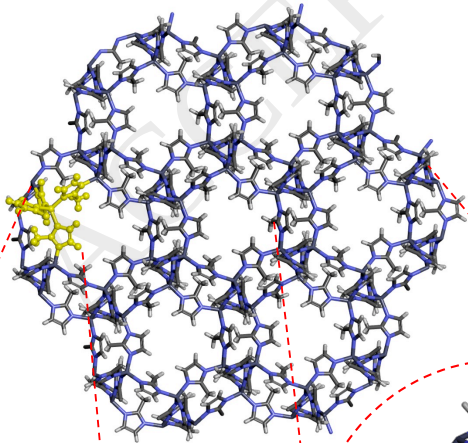


MOFs

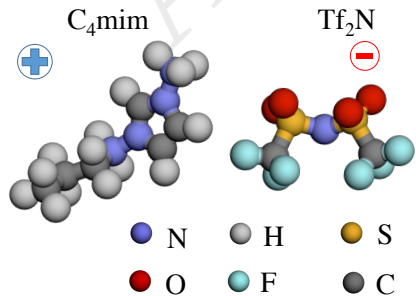
Carbon  
nanotubesMesoporous  
silicasCO<sub>2</sub>0.12  
nm

AC

PT







(a)

